Design Verification Plan & Report - VBF-T6LC-DVPR-01

Design Verification Plan & Report (virtual-test) - VBF-T6LC-DVPR-01

	Condition	Result value	Determination	Source
e capacity	1C CC discharge, 298.15 K, DFN	4.804 Ah	PASS (>= nominal 4.5 Ah)	`cell/final_1c_dfn_h45.json:capacity_ah`
energy density	contract-caliber stack formula	1005.8 Wh/L	PASS (>= 950 Wh/L)	`cell/final_1c_energy_dfn_h45.json:energy_d
teau	discharge-time midpoint	4.115 V	PASS (>= 4.1 V)	`cell/final_1c_energy_dfn_h45.json:midpoint_
rge temperature rise	4C charge, 45 degC , lumped thermal, DFN	322.59 K (49.44 degC)	PASS (<= 323.15 K / 50 degC)	`cell/final_4c_safety_h45_dfn.json:T_max_K`
rge plating	`--plating`, anode surface potential	min +0.0894 V	PASS (>= 0 V, no plating)	`cell/final_4c_safety_h45_dfn.json:anode_pot
hickness after 100 cycles	1C aging 100 cycles, DFN (and SPMe)	9.5 nm (DFN) / 233 nm (SPMe)	PASS (<= 500 nm)	`cell/final_aging_dfn_h45.json:sei_thickness_
dow	parameter set	2.5 - 4.7 V	PASS (system definition)	`data/LNMO.json`

Items explicitly N/A (beyond pure-simulation boundary, require physical experiment): nail penetration, overcharge-to-thermal-runaway, crush, drop, rate-pulse internal resistance, long-term calendar life.

Conclusion

All six simulated verification items PASS against the entry-0 criteria. Uncovered physical-abuse items are listed above and are outside the virtual-test boundary.